

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 7: Ratio and Proportion

Previous year board questions, topic-wise: 8 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 Properties of proportion

1. Use the properties of proportion to solve  $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$ , where  $x \neq 0$ .  
ICSE 2025, 3 marks
2. Given  $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$ , find (a)  $\frac{a+b}{a-b}$  (b) the ratio  $a : b$ , using the properties of proportion.  
ICSE 2024, 4 marks
3. Use componendo and dividendo to solve  $\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$ . ICSE 2023, 3 marks
4. Find  $x : y$ , given  $\frac{x^2 + 2x}{2x + 4} = \frac{y^2 + 3y}{3y + 9}$ . ICSE 2020, 3 marks
5. If  $x = \frac{\sqrt{2a+1} + \sqrt{2a-1}}{\sqrt{2a+1} - \sqrt{2a-1}}$ , prove that  $x^2 - 4ax + 1 = 0$ . ICSE 2020, 4 marks

## 2 Proportion

6. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in  $x$  s and  $y$  m in 5 s. Find  $x$  and  $y$ . (a)  $x = 4, y = 150$  (b)  $x = 3, y = 100$  (c)  $x = 4, y = 100$  (d)  $x = 3, y = 150$  ICSE 2024, 1 mark
7. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion? ICSE 2023, 3 marks

## 3 Continued proportion

8. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 ICSE 2023, 1 mark

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